# MA1101: Mathematics-1

## Tutorial Sheets 2, 3 & 4 Complete Solutions Manual

**Comprehensive Exam-Ready Solutions with Explanations & Key Points**

October 27, 2026

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# 1 Tutorial Sheet 2: Integration, Polar Coordinates & FTC

## Problem 1: Piecewise Integral & Graph

**Problem 1**

Compute the integral $\int_0^1 f(x)dx$ where

$$f(x) = \begin{cases} x & \text{if } 0 \le x \le c \\ c\frac{1-x}{1-c} & \text{if } c \le x \le 1 \end{cases}$$

where $c$ is a fixed real number, $0 < c < 1$.

**Solution:** We split the integral at $x = c$:

$$\int_0^1 f(x)dx = \int_0^c x\,dx + \int_c^1 c\frac{1-x}{1-c}\,dx$$

1. First integral:

$$\int_0^c x\,dx = \left[\frac{x^2}{2}\right]_0^c = \frac{c^2}{2}$$

2. Second integral: Let $u = 1 - x$, $du = -dx$. When $x = c, u = 1 - c$; when $x = 1, u = 0$:

$$\int_c^1 c\frac{1-x}{1-c}\,dx = \frac{c}{1-c}\int_{1-c}^0 (-u)\,du = \frac{c}{1-c}\int_0^{1-c} u\,du = \frac{c}{1-c}\left[\frac{u^2}{2}\right]_0^{1-c} = \frac{c(1-c)}{2}$$

3. Combining both parts:

$$\int_0^1 f(x)dx = \frac{c^2}{2} + \frac{c-c^2}{2} = \frac{c}{2}$$

**Exam Pointer & Geometric Interpretation**

The graph of $f(x)$ forms a triangle with vertices at $(0,0)$, $(c,c)$, and $(1,0)$. The area of this triangle is $\frac{1}{2} \times$ base $\times$ height $= \frac{1}{2} \times 1 \times c = \frac{c}{2}$. Always check integration results against basic geometry when linear functions form simple polygons!

## Problem 2: Area Between Curves

**Problem 2(i)**

Compute the area of the region $S$ between $f(x) = |x|$ and $g(x) = x^2 - 1$ over $a = -1, b = 1$.

**Solution:** On $[-1, 1]$, $|x| \geq 0$ while $x^2 - 1 \leq 0$, so $f(x) \geq g(x)$ throughout $[-1, 1]$. By symmetry, the area $A$ is:

$$A = \int_{-1}^{1} (|x| - (x^2 - 1))\, dx = 2\int_{0}^{1} (x - x^2 + 1)\, dx$$

$$A = 2\left[\frac{x^2}{2} - \frac{x^3}{3} + x\right]_0^1 = 2\left(\frac{1}{2} - \frac{1}{3} + 1\right) = 2\left(\frac{7}{6}\right) = \frac{7}{3}$$

**Problem 2(ii)**

Compute the area between $f(x) = |x| + |x - 1|$ and $g(x) = 0$ over $[-1, 2]$.

**Solution:** We express $f(x)$ piecewise across intervals bounded by $x = 0$ and $x = 1$:

$$f(x) = \begin{cases} -x - (x-1) = 1 - 2x & \text{if } x < 0 \\ x - (x-1) = 1 & \text{if } 0 \leq x \leq 1 \\ x + (x-1) = 2x - 1 & \text{if } x > 1 \end{cases}$$

We split the integral $\int_{-1}^{2} f(x)dx$:

$$\int_{-1}^{0} (1 - 2x)dx = \left[x - x^2\right]_{-1}^{0} = 0 - (-1 - 1) = 2$$

$$\int_{0}^{1} 1\, dx = 1$$

$$\int_{1}^{2} (2x - 1)dx = \left[x^2 - x\right]_{1}^{2} = (4 - 2) - (1 - 1) = 2$$

Total Area $= 2 + 1 + 2 = 5$.

## Problem 3: Polar Coordinate Areas

**Problem 3(i), (ii), (iii)**

Compute the area of the radial set for:

1. $f(\theta) = 4\sin\theta, 0 \leq \theta \leq \pi$
2. $f(\theta) = \sqrt{|\cos\theta|}, 0 \leq \theta \leq 2\pi$
3. $f(\theta) = |\sin 2\theta|, 0 \leq \theta \leq 2\pi$

**Solutions:** The area formula in polar coordinates is $A = \frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$.

1. For $r = 4\sin\theta$:

$$A = \frac{1}{2}\int_0^{\pi} (4\sin\theta)^2 d\theta = 8\int_0^{\pi} \sin^2\theta\, d\theta = 8\int_0^{\pi} \frac{1 - \cos 2\theta}{2}\, d\theta = 4\pi$$

*(Note: $r = 4\sin\theta$ is a circle of radius 2 centered at $(0, 2)$, area $\pi(2)^2 = 4\pi$.)*

2. For $r = \sqrt{|\cos\theta|}$: By 4-fold symmetry across quadrants:

$$A = 4 \cdot \frac{1}{2} \int_0^{\pi/2} |\cos\theta|\, d\theta = 2 \int_0^{\pi/2} \cos\theta\, d\theta = 2[\sin\theta]_0^{\pi/2} = 2$$

3. For $r = |\sin 2\theta|$ (Four-leaved rose): By 8-fold symmetry across each half-petal ($0 \le \theta \le \pi/4$):

$$A = 8 \cdot \frac{1}{2} \int_0^{\pi/4} \sin^2 2\theta\, d\theta = 4 \int_0^{\pi/4} \frac{1-\cos 4\theta}{2}\, d\theta = 2\left[\theta - \frac{\sin 4\theta}{4}\right]_0^{\pi/4} = 2\left(\frac{\pi}{4}\right) = \frac{\pi}{2}$$

## Problem 4: Volume of Revolution

**Problem 4(i), (ii)**

Compute volume revolving $y = f(x)$ about the x-axis: (i) $f(x) = x^2, -1 \le x \le 2$. (ii) $f(x) = \sin x + \cos x, 0 \le x \le \pi$.

**Solutions:** Using the disk method $V = \pi \int_a^b [f(x)]^2 dx$:

1. $V = \pi \int_{-1}^{2} (x^2)^2 dx = \pi \int_{-1}^{2} x^4 dx = \pi \left[\frac{x^5}{5}\right]_{-1}^{2} = \frac{\pi}{5}(32 - (-1)) = \frac{33\pi}{5}$.
2. $[f(x)]^2 = (\sin x + \cos x)^2 = \sin^2 x + \cos^2 x + 2\sin x \cos x = 1 + \sin 2x$.

$$V = \pi \int_0^{\pi} (1 + \sin 2x) dx = \pi \left[x - \frac{\cos 2x}{2}\right]_0^{\pi} = \pi\left(\left(\pi - \frac{1}{2}\right) - \left(0 - \frac{1}{2}\right)\right) = \pi^2$$

## Problem 5 & 6: Average Value of Functions

**Problem 5 & 6**

5(i) Average of $f(x) = x^{1/2}$ on $[0, 4]$.
5(ii) Average of $f(x) = \sin x \cos x$ on $[0, \pi/4]$.
6(i) Find $c \in (0, a)$ such that $f(c) = A(f)$ for $f(x) = x^n$ on $[0, a]$.

**Solutions:** The average value $A(f) = \frac{1}{b-a} \int_a^b f(x) dx$.

5(i) $A(f) = \frac{1}{4} \int_0^4 x^{1/2} dx = \frac{1}{4} \left[\frac{2}{3} x^{3/2}\right]_0^4 = \frac{1}{4} \cdot \frac{2}{3} \cdot 8 = \frac{4}{3}$.

5(ii) $A(f) = \frac{1}{\pi/4} \int_0^{\pi/4} \frac{1}{2} \sin 2x\, dx = \frac{4}{\pi} \cdot \frac{1}{2} \left[-\frac{\cos 2x}{2}\right]_0^{\pi/4} = \frac{2}{\pi}\left(0 + \frac{1}{2}\right) = \frac{1}{\pi}$.

6(ii) For $f(x) = x^n$ on $[0, a]$:

$$A(f) = \frac{1}{a} \int_0^a x^n dx = \frac{1}{a} \frac{a^{n+1}}{n+1} = \frac{a^n}{n+1}$$

Set $f(c) = c^n = \frac{a^n}{n+1} \implies c = \frac{a}{(n+1)^{1/n}}$. For $n = 2$, $c = \frac{a}{\sqrt{3}}$.

## Problems 7–14: Fundamental Theorem of Calculus

**Problem 7**

Area between $f(x) = x^2$ and $g(x) = cx^3$ ($c > 0$) over $[0, 1/c]$ is $2/3$. Find $c$.

**Solution:** On $[0, 1/c]$, $x^2 \ge cx^3$. Area $A = \int_0^{1/c} (x^2 - cx^3) dx = \left[\frac{x^3}{3} - \frac{cx^4}{4}\right]_0^{1/c} = \frac{1}{3c^3} - \frac{1}{4c^3} = \frac{1}{12c^3}$.
Set $\frac{1}{12c^3} = \frac{2}{3} \implies 24c^3 = 3 \implies c^3 = \frac{1}{8} \implies c = \frac{1}{2}$.

**Problem 8**

Given $f(\pi) = 2$ and $\int_0^{\pi} [f(x) + f''(x)] \sin x\, dx = 5$. Compute $f(0)$.

**Solution:** Use integration by parts on $\int_0^\pi f''(x)\sin x\,dx$:

$$\int_0^\pi f''(x)\sin x\,dx = \left[f'(x)\sin x\right]_0^\pi - \int_0^\pi f'(x)\cos x\,dx = 0 - \left(\left[f(x)\cos x\right]_0^\pi - \int_0^\pi f(x)(-\sin x)dx\right)$$

$$= -\left(f(\pi)(-1) - f(0)(1) + \int_0^\pi f(x)\sin x\,dx\right) = f(\pi) + f(0) - \int_0^\pi f(x)\sin x\,dx$$

Substitute back into integral:

$$\int_0^\pi f(x)\sin x\,dx + \left(f(\pi) + f(0) - \int_0^\pi f(x)\sin x\,dx\right) = 5 \implies f(\pi) + f(0) = 5$$

Since $f(\pi) = 2$, we have $2 + f(0) = 5 \implies f(0) = 3$.

## Problem 9

Compute $f'(x)$ for $f(x) = \int_{x^3}^{x^2} \frac{t^6}{1+t^4}dt$.

**Solution:** By Leibniz's Rule $\frac{d}{dx}\int_{a(x)}^{b(x)} g(t)dt = g(b(x))b'(x) - g(a(x))a'(x)$:

$$f'(x) = \frac{(x^2)^6}{1+(x^2)^4}\cdot(2x) - \frac{(x^3)^6}{1+(x^3)^4}\cdot(3x^2) = \frac{2x^{13}}{1+x^8} - \frac{3x^{20}}{1+x^{12}}$$

## Problem 10

Let $f(x) = \int_x^2 \frac{1}{\sqrt{1+t^3}}dt$. Find $\int_0^2 xf(x)dx$.

**Solution:** Use integration by parts: $u = f(x)$, $dv = x\,dx \implies du = f'(x)dx = -\frac{1}{\sqrt{1+x^3}}dx$, $v = \frac{x^2}{2}$.

$$\int_0^2 xf(x)dx = \left[\frac{x^2}{2}f(x)\right]_0^2 - \int_0^2 \frac{x^2}{2}\left(-\frac{1}{\sqrt{1+x^3}}\right)dx = \left(\frac{4}{2}f(2) - 0\right) + \frac{1}{2}\int_0^2 \frac{x^2}{\sqrt{1+x^3}}dx$$

Note $f(2) = \int_2^2 \cdots = 0$. For the remaining integral, let $w = 1+x^3, dw = 3x^2dx$:

$$= \frac{1}{2}\cdot\frac{1}{3}\int_1^9 w^{-1/2}dw = \frac{1}{6}\left[2\sqrt{w}\right]_1^9 = \frac{1}{3}(3-1) = \frac{2}{3}$$

## Problem 11

Show $\int_0^\pi xf(\sin x)dx = \frac{\pi}{2}\int_0^\pi f(\sin x)dx$ and deduce $\int_0^\pi \frac{x\sin x}{1+\cos^2 x}dx = \frac{\pi^2}{4}$.

**Solution:** (i) Let $I = \int_0^\pi xf(\sin x)dx$. Substitute $u = \pi - x, dx = -du$:

$$I = \int_0^\pi (\pi-u)f(\sin(\pi-u))du = \pi\int_0^\pi f(\sin u)du - I \implies 2I = \pi\int_0^\pi f(\sin x)dx \implies I = \frac{\pi}{2}\int_0^\pi f(\sin x)dx$$

(ii) Apply this identity to $f(\sin x) = \frac{\sin x}{1+\cos^2 x}$:

$$I = \frac{\pi}{2}\int_0^\pi \frac{\sin x}{1+\cos^2 x}dx$$

Let $t = \cos x, dt = -\sin x dx$:

$$I = \frac{\pi}{2}\int_{-1}^1 \frac{dt}{1+t^2} = \pi\int_0^1 \frac{dt}{1+t^2} = \pi\left[\arctan t\right]_0^1 = \pi\left(\frac{\pi}{4}\right) = \frac{\pi^2}{4}$$

**Problem 12, 13, 14**

12. $g(1) = 1, g'(x^2) = x^3$. Find $g(4)$.
13. Inverse $g = f^{-1}$ of $f(x) = \int_0^x (1+t^3)^{-1/2} dt$. Find $g''(y)$.
14(ii) Derivative of $f(x) = \int_1^{\int_1^x t^2 dt} \frac{1}{t} dt$.

**Solutions:** 12. Let $u = x^2 \implies x = \sqrt{u}$. Then $g'(u) = u^{3/2}$. $g(4) - g(1) = \int_1^4 g'(u)du = \int_1^4 u^{3/2}du = \left[\frac{2}{5}u^{5/2}\right]_1^4 = \frac{2}{5}(32-1) = \frac{62}{5}$. Since $g(1) = 1$, $g(4) = 1 + \frac{62}{5} = \frac{67}{5}$.

13. $f'(x) = (1+x^3)^{-1/2} > 0 \implies f$ strictly increasing. $g = f^{-1} \implies g'(y) = \frac{1}{f'(x)} = \sqrt{1+x^3} = \sqrt{1+g(y)^3}$. Differentiating w.r.t $y$:

$$g''(y) = \frac{1}{2\sqrt{1+g(y)^3}} \cdot 3g(y)^2 g'(y) = \frac{3g(y)^2\sqrt{1+g(y)^3}}{2\sqrt{1+g(y)^3}} = \frac{3}{2}g(y)^2$$

Thus $g''(y)$ is proportional to $g(y)^2$ with constant $c = 3/2$.

14(ii) Let $h(x) = \int_1^x t^2 dt = \frac{x^3-1}{3}$. Then $f(x) = \int_1^{h(x)} \frac{1}{t} dt = \ln(h(x))$. By Chain Rule / Leibniz Rule:

$$f'(x) = \frac{1}{h(x)} \cdot h'(x) = \frac{x^2}{\frac{x^3-1}{3}} = \frac{3x^2}{x^3-1}$$

# 2 Tutorial Sheet 3: Sequences, Series & Ordinary Differential Equations

## Problem 1: Sequence Convergence

**Problem 1(a)–(d)**

Find limits: (a) $u_n = \frac{n}{n+1} - \frac{n+1}{n}$. (b) $u_n = \frac{1+(-1)^n}{n}$.
(c) $u_n = \sqrt{n+1} - \sqrt{n}$. (d) $u_n = e^{-in\pi/2}$.

**Solutions:** (a) $u_n = \frac{n^2-(n+1)^2}{n(n+1)} = \frac{-2n-1}{n^2+n} \to 0$ as $n \to \infty$. Converges to 0.
(b) $-\frac{2}{n} \le \frac{1+(-1)^n}{n} \le \frac{2}{n}$. By Squeeze Theorem, $\lim u_n = 0$. Converges to 0.
(c) $u_n = \frac{(n+1)-n}{\sqrt{n+1}+\sqrt{n}} = \frac{1}{\sqrt{n+1}+\sqrt{n}} \to 0$. Converges to 0.
(d) $u_n \in \{-i, -1, i, 1\}$ periodically. Oscillates without a single limit. Diverges.

## Problem 2 & 3: Series Convergence

**Problem 2(a)–(d) & 3(a)–(c)**

Find sums of convergent series and test conditional/absolute convergence.

**Solutions:** 2(a) Geometric series $\sum_{n=1}^{\infty} 2(1/3)^{n-1} = \frac{a}{1-r} = \frac{2}{1-1/3} = 3$.
2(b) Telescoping: $\frac{1}{n^2-1} = \frac{1}{2}\left(\frac{1}{n-1} - \frac{1}{n+1}\right)$. Sum $= \frac{1}{2}((1+1/2)) = \frac{3}{4}$.
2(c) Telescoping: $\frac{2n+1}{n^2(n+1)^2} = \frac{1}{n^2} - \frac{1}{(n+1)^2}$. Sum $= \sum_{n=1}^{\infty}\left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right) = 1$.
3(a) $\sum \frac{(-1)^{n(n-1)/2}}{2^n}$: $|a_n| = \frac{1}{2^n}$. Sum of absolute values converges (geometric $r = 1/2$). Absolutely convergent.
3(b) $\sum \frac{(-1)^n}{n^{3/2}}$: Absolute series $\sum \frac{1}{n^{3/2}}$ is a $p$-series with $p = 3/2 > 1$, which converges. Absolutely convergent.
3(c) $\lim_{n\to\infty} \frac{n^2}{n^2+1} = 1 \neq 0$. Fails the $n$-th term test for convergence. Diverges.

## Problems 4–8: Differential Equations

**Problem 4 & 5**

4. Curve through $(1, 0)$ with slope $dy/dx = -x/y$.
5(a) IVP: $y' - 2y = e^{2x}, y(0) = 3$. (b) IVP: $y' - \frac{2}{x}y = x, y(1) = 2$.

**Solutions:** 4. $y\,dy = -x\,dx \implies \frac{y^2}{2} = -\frac{x^2}{2} + C \implies x^2 + y^2 = 2C$. Passing through $(1, 0) \implies 1 + 0 = 2C \implies x^2 + y^2 = 1$ (Unit circle).
5(a) Integrating factor $I(x) = e^{\int -2dx} = e^{-2x}$. $\frac{d}{dx}(ye^{-2x}) = e^{-2x}e^{2x} = 1 \implies ye^{-2x} = x + C \implies y(x) = (x + C)e^{2x}$. $y(0) = 3 \implies C = 3 \implies y(x) = (x + 3)e^{2x}$.
5(b) $I(x) = e^{\int -2/x dx} = x^{-2}$. $\frac{d}{dx}(yx^{-2}) = x \cdot x^{-2} = \frac{1}{x} \implies yx^{-2} = \ln x + C \implies y(x) = x^2 \ln x + Cx^2$. $y(1) = 2 \implies 0 + C = 2 \implies y(x) = x^2 \ln x + 2x^2$.

**Problem 7: Reduction of Order**

Find second solution using $y_2(x) = v(x)y_1(x)$: 7(a) $x^2y'' + xy' - y = 0, y_1(x) = x$.

**Solution:** Standard form $y'' + \frac{1}{x}y' - \frac{1}{x^2}y = 0$. Abel's formula / reduction formula: $v'(x) = \frac{1}{y_1^2}e^{-\int P(x)dx} = \frac{1}{x^2}e^{-\ln x} = \frac{1}{x^3}$. Integrating: $v(x) = \int x^{-3}dx = -\frac{1}{2x^2}$. Thus $y_2(x) = v(x)y_1(x) = x\left(-\frac{1}{2x^2}\right) = -\frac{1}{2x}$, or simply $y_2(x) = x^{-1}$. General Solution: $y(x) = c_1x + c_2x^{-1}$.

**Problem 8: Variation of Parameters**

Solve $(D^2 + 1)y = \csc x$.

**Solution:** 1. Complementary function: $y_c = c_1\cos x + c_2\sin x$. $y_1 = \cos x, y_2 = \sin x$. Wronskian $W(y_1, y_2) = \cos x(\cos x) - \sin x(-\sin x) = 1$. 2. Particular solution $y_p = u_1y_1 + u_2y_2$:

$$u_1 = -\int \frac{y_2 g(x)}{W}dx = -\int \sin x \csc x\, dx = -\int 1\, dx = -x$$

$$u_2 = \int \frac{y_1 g(x)}{W}dx = \int \cos x \csc x\, dx = \int \cot x\, dx = \ln|\sin x|$$

3. $y_p(x) = -x\cos x + \sin x \ln|\sin x|$. General Solution: $y(x) = c_1\cos x + c_2\sin x - x\cos x + \sin x \ln|\sin x|$.

# 3 Tutorial Sheet 4: Multivariable Calculus & Line Integrals

## Problems 1–3: Multivariable Limits & Continuity

**Problem 1**

Let $f(x,y) = \frac{x^2y^2}{x^2y^2+(x-y)^2}$. Show iterated limits are 0, but total limit as $(x,y) \to (0,0)$ does not exist.

**Solution:** 1. Iterated limits:

$$\lim_{x\to 0}\left[\lim_{y\to 0}\frac{x^2y^2}{x^2y^2+(x-y)^2}\right] = \lim_{x\to 0}\left[\frac{0}{0+x^2}\right] = 0$$

$$\lim_{y\to 0}\left[\lim_{x\to 0}\frac{x^2y^2}{x^2y^2+(x-y)^2}\right] = \lim_{y\to 0}\left[\frac{0}{0+y^2}\right] = 0$$

2. Path $y = x$:

$$f(x,x) = \frac{x^4}{x^4+0} = 1 \implies \lim_{x\to 0} f(x,x) = 1 \neq 0$$

Since limits along different paths ($y = 0$ vs $y = x$) yield different values (0 vs 1), the limit as $(x,y) \to (0,0)$ does not exist.

**Problem 7: Directional Derivative**

Evaluate directional derivative of $f(x,y,z) = x^2+2y^2+3z^2$ at $(1,1,0)$ in direction $\mathbf{v} = \mathbf{i}-\mathbf{j}+2\mathbf{k}$.

**Solution:** 1. Gradient $\nabla f = (2x)\mathbf{i} + (4y)\mathbf{j} + (6z)\mathbf{k}$. At $(1,1,0)$, $\nabla f(1,1,0) = 2\mathbf{i} + 4\mathbf{j} + 0\mathbf{k}$. 2. Unit vector $\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\mathbf{i}-\mathbf{j}+2\mathbf{k}}{\sqrt{1^2+(-1)^2+2^2}} = \frac{1}{\sqrt{6}}(\mathbf{i}-\mathbf{j}+2\mathbf{k})$. 3. $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = (2\mathbf{i}+4\mathbf{j}) \cdot \frac{1}{\sqrt{6}}(\mathbf{i}-\mathbf{j}+2\mathbf{k}) = \frac{2(1)+4(-1)+0(2)}{\sqrt{6}} = \frac{-2}{\sqrt{6}} = -\frac{\sqrt{6}}{3}$.

## Problems 10–15: Optimization & Lagrange Multipliers

**Problem 10(i)**

Find stationary points of $z = x^3 + y^3 - 3xy$.

**Solution:** 1. Partial derivatives: $f_x = 3x^2 - 3y = 0 \implies y = x^2$. $f_y = 3y^2 - 3x = 0 \implies x = y^2$. 2. Substituting $y = x^2$: $x = (x^2)^2 = x^4 \implies x(x^3-1) = 0 \implies x = 0$ or $x = 1$. Stationary points: $(0,0)$ and $(1,1)$. 3. Second derivatives: $f_{xx} = 6x, f_{yy} = 6y, f_{xy} = -3$. Discriminant $D = f_{xx}f_{yy} - (f_{xy})^2 = 36xy - 9$. * At $(0,0)$: $D = -9 < 0 \implies (\mathbf{0},\mathbf{0})$ is a Saddle Point. * At $(1,1)$: $D = 36 - 9 = 27 > 0$ and $f_{xx}(1,1) = 6 > 0 \implies (\mathbf{1},\mathbf{1})$ is a Local Minimum with $z = -1$.

**Problem 13: Lagrange Multipliers**

Find extreme values of $f(x,y,z) = x - 2y + 2z$ on $x^2+y^2+z^2 = 1$.

**Solution:** 1. Constraint $g(x,y,z) = x^2+y^2+z^2-1 = 0$. 2. $\nabla f = \lambda \nabla g \implies (1,-2,2) = \lambda(2x,2y,2z)$.

$$x = \frac{1}{2\lambda}, \quad y = -\frac{1}{\lambda}, \quad z = \frac{1}{\lambda}$$

3. Substitute into $x^2+y^2+z^2 = 1$:

$$\frac{1}{4\lambda^2} + \frac{1}{\lambda^2} + \frac{1}{\lambda^2} = 1 \implies \frac{9}{4\lambda^2} = 1 \implies \lambda = \pm\frac{3}{2}$$

* For $\lambda = 3/2$: $(x,y,z) = (1/3,-2/3,2/3)$, $f = 1/3 - 2(-2/3) + 2(2/3) = \mathbf{3}$ (Maximum). * For $\lambda = -3/2$: $(x,y,z) = (-1/3,2/3,-2/3)$, $f = -1/3 - 2(2/3) + 2(-2/3) = \mathbf{-3}$ (Minimum).

## Problems 16–18: Line Integrals & Conservative Fields

**Problem 17(i)**

Compute $\int_C (x^2 - 2xy)dx + (y^2 - 2xy)dy$ along parabola $y = x^2$ from $(-2, 4)$ to $(1, 1)$.

**Solution:** Parameterize using $x = t, y = t^2, dy = 2t\,dt$ for $t \in [-2, 1]$:

$$\int_{-2}^{1} \left((t^2 - 2t^3) + (t^4 - 2t^3)(2t)\right) dt = \int_{-2}^{1} (t^2 - 2t^3 + 2t^5 - 4t^4)dt$$

$$\left[\frac{t^3}{3} - \frac{t^4}{2} + \frac{t^6}{3} - \frac{4t^5}{5}\right]_{-2}^{1}$$

Evaluating at $t = 1$: $\frac{1}{3} - \frac{1}{2} + \frac{1}{3} - \frac{4}{5} = \frac{2}{3} - \frac{13}{10} = -\frac{19}{30}$. Evaluating at $t = -2$: $-\frac{8}{3} - 8 + \frac{64}{3} + \frac{128}{5} = \frac{56}{3} - 8 + \frac{128}{5} = \frac{32}{3} + \frac{128}{5} = \frac{544}{15}$. Value $= -\frac{19}{30} - \frac{1088}{30} = -\frac{\mathbf{1107}}{\mathbf{30}} = -\frac{\mathbf{369}}{\mathbf{10}} = \mathbf{-36.9}$.

**Problem 18(i)**

Test if $\mathbf{f}(x, y) = (2xe^y + y)\mathbf{i} + (x^2e^y + x - 2y)\mathbf{j}$ is conservative and find $\phi$.

**Solution:** 1. Let $P = 2xe^y + y$ and $Q = x^2e^y + x - 2y$.

$$\frac{\partial P}{\partial y} = 2xe^y + 1, \quad \frac{\partial Q}{\partial x} = 2xe^y + 1$$

Since $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, $\mathbf{f}$ is conservative. 2. Potential function $\phi(x, y)$:

$$\phi(x, y) = \int (2xe^y + y)dx = x^2e^y + xy + g(y)$$

$$\frac{\partial \phi}{\partial y} = x^2e^y + x + g'(y) = Q(x, y) = x^2e^y + x - 2y \implies g'(y) = -2y \implies g(y) = -y^2 + C$$

$$\phi(\mathbf{x}, \mathbf{y}) = \mathbf{x^2e^y + xy - y^2 + C}$$